Kanokpon Mettasat

Electrical & Computer Engineering, Institute Honors Program -- post-quantum cryptography & blockchain security research

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ABOUT

ECE cadet at Virginia Military Institute (Class of 2027, Institute Honors Program), applying to MS programs in Electrical Engineering for Fall 2027: secure and embedded systems, post-quantum cryptographic engineering for resource-constrained devices. Undergraduate researcher on quantum-resistant blockchain security (first-author paper accepted at IEEE UEMCON 2026; second published in Computers, MDPI). Hardware lead on a fabricated KiCad PCB; team lead at IEEE SoutheastCon 2026 (top seven of forty); sole developer of Officer Parker, a real-time voice agent. President, Eta Kappa Nu (IEEE-HKN), VMI, 2026-27.

EDUCATION

Virginia Military Institute -- BS, Electrical and Computer Engineering (Aug 2023 -- May 2027 (expected))

- Institute Honors Program; Dean's List; Overall GPA 3.81 / Major GPA 4.00.
- President, Eta Kappa Nu (IEEE-HKN), VMI chapter (2026 -- 2027); Tau Beta Pi, Virginia Chapter (Nov 2025).
- IEEE SoutheastCon 2026 -- top 7 of 40 teams, Undergraduate Circuit Design (Mar 2026).
- Chulachomklao Royal Military Academy (CRMA), Class 74 -- selected first of the class for the Royal Thai Army's overseas study programme to the United States (2022).

PUBLICATIONS

- K. Mettasat and E. Fathalla. “Post-Quantum Cryptography for Blockchain: A Systematic Mapping by Cryptographic Family.” *2026 IEEE 17th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), Armonk, New York*, Oct 2026. Accepted and presented; to appear in IEEE Xplore.
- K. Mettasat, E. Fathalla and M. Azab. “A Systematic Mapping of Quantum Threats and Defenses for Blockchain.” *Computers (MDPI), vol. 15, no. 11, art. 743*, Oct 2026. Published; open access.

Talks (presenter): IEEE UEMCON 2026, Armonk, NY (Oct 2026); SURF 2026, Wofford College (10 Oct 2026).

PROJECTS

SURI -- Systematic mapping of quantum-resistant blockchain security.
Undergraduate research with Dr. Efat Fathalla (VMI ECE): systematic mapping of 387 studies under a PRISMA 2020 protocol; idempotent Python pipeline rebuilds the corpus from the search log.

IEEE Xplore + MDPI | PRISMA 2020 | Evidence-maturity scale (L0-L5) | Multi-label coding

Motion-triggered novelty device (EE-382) -- schematic to fabricated PCB.
Hardware lead, EE-382 (Spring 2026): sole designer of the electronics -- KiCad schematic and PCB, fabricated and hand-assembled; Arduino-class firmware fusing light, sound and motion sensing to trigger audio; debugged on the oscilloscope.

KiCad (schematic + PCB) | Arduino-class microcontroller | Light / sound / motion sensors | Audio playback

Officer Parker -- real-time voice agent for mock US visa interviews.
Browser voice agent: 24 kHz PCM capture, speech-to-speech WebSocket session, structured report from the transcript, grounded in the applicant's own DS-160 so contradictions surface. Sole developer; AssemblyAI Voice Agent Hackathon (Sep 2026); MIT-licensed, in production.

JavaScript | Web Audio (AudioWorklet, 24 kHz PCM) | WebSocket speech-to-speech | Vercel serverless functions

SKILLS

Languages: Python | Hardware: KiCad (schematic to fabricated PCB) | Engineering: Digital systems design | Research: Systematic review methodology | Tooling: Git | Web / Infrastructure: FastAPI

Live portfolio: km-portfolio.onrender.com/portfolio